

InsightIQ — Mathematical Framework for Business Optimization

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STAGE 01

Data Layer

Raw business records collected and cleaned

Step 1 · Business data acquisition

Historical sales, inventory levels, supplier catalogues (cost, capacity, lead time) and customer demand records are ingested from the organisation's BI sources.

Sales history

Inventory

Supplier data

Customer data

Step 2 · Preprocessing & structuring

Missing values handled, series aligned to time periods, demand discretised into states, and observation weights assigned (recent periods weighted higher).

STAGE 02

Prediction Layer

Demand estimated point-wise and probabilistically

Step 3 · Demand forecasting

WEIGHTED LEAST SQUARES

Fits a trend to weighted historical sales to produce the expected demand for the next period.

Step 4 · Uncertainty modelling

MARKOV CHAINS

Transition probabilities between demand states (low / medium / high) give the probability distribution of future demand.

Merge → **expected demand D*** with an associated uncertainty profile, passed forward to optimization.

STAGE 03

Optimization Layer

Suppliers chosen, optimality proven, price maximised

Step 5 · Supplier selection

CONSTRAINED CONVEX OPTIMIZATION

Minimises total procurement cost of meeting forecast demand D* across candidate suppliers.

cost

capacity

inventory

delivery

budget

Step 6 · Optimality check

KKT CONDITIONS

Stationarity, primal & dual feasibility and complementary slackness are verified for the candidate solution.

satisfied → proceed to Step 7 not satisfied → return to Step 5

Step 7 · Price optimization

GRADIENT-BASED OPTIMIZATION

Iteratively ascends the profit function — using predicted demand and the optimal procurement cost from Step 5 — until the selling price converges.

STAGE 04

Decision Output

Unified, justified recommendations

Step 8 · Mathematical Decision Engine output

FORECAST

Expected demand per product & period

SOURCING

Optimal supplier mix & order quantities

PRICING

Profit-maximising selling price

JUSTIFICATION

KKT-verified proof of optimality

Feedback loop → realised sales are appended to the data layer, re-weighting Step 2 and refreshing every downstream model.